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# DEMAND, INFORMATION, AND COMPETITION: WHY DO FOOD PRICES FALL AT SEASONAL DEMAND PEAKS?*

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Prices for seasonal food products fall at demand peaks. Price declines are not driven by falling agricultural input prices; indeed, farm to retail margins narrow sharply. I use electronic scanner data from a sample of US supermarkets to show that seasonal price declines are closely linked to market concentration, and are much larger in markets with several rivals than where a single brand dominates. Seasonal demand increases reduce the effective costs of informative advertising, and increased informative advertising by retailers and manufacturers in turn may allow for increased market information and greater price sensitivity on the part of buyers.

## I. INTRODUCTION

Many food products have distinctive seasonal demand patterns. For example, sales of flour and baking supplies surge in late fall, shortly before the Christmas holidays, while sales of barbecue sauces, ketchup, and hot dogs peak in early summer. This paper investigates a striking empirical regularity among food products with strong seasonal demand patterns: prices move counter-cyclically, falling at seasonal quantity peaks, and rising off-peak as demand falls.

Figure 1 summarizes the pattern for supermarket sales of one product, all-purpose wheat flour, for the period from January 1989 through December 1994, using monthly price and quantity indexes (with means set to 100). Quantity peaks sharply late in the fall, rising to as much as five times that of off-peak months. Prices (the solid line) show much less seasonal variation but nevertheless clearly move countercyclically. Retail flour prices rise sharply in off-peak (summer) months, and persistently fall during peak consumption months.

Bryan and Cecchetti [1995] opine that supply fluctuations may explain much of the seasonal fluctuation that they observe in retail food prices. But the primary agricultural commodity used in all-purpose flour is winter

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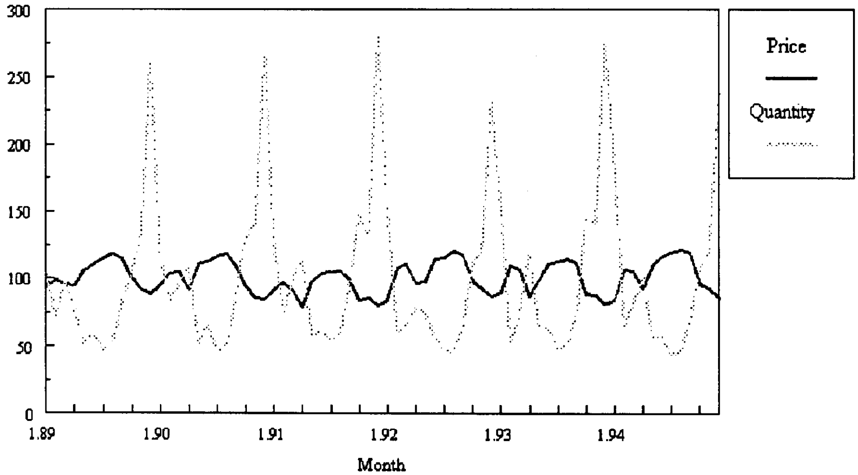


Figure 1
Monthly Flour Price and Quantity Movements

wheat, which is harvested in the spring. Wheat prices show some modest seasonal fluctuations, but hit seasonal highs in the late fall, just when retail flour prices hit troughs.

Countercyclical seasonal price movements aren't completely unexpected. There is a growing literature that seeks to document and explain margins (ratios of price to marginal cost) that move counter to business cycles (summarized in Rotemberg and Woodford [1999]). Models in that literature explain how shifts in demand might alter tacitly collusive pricing agreements among oligopolists, leading to prices that move with demand cycles.

The analyses reported in this paper use price data for 48 food products with strong seasonal sales patterns. Almost all match figure 1 in three key respects: (1) prices show much smaller seasonal fluctuations than quantities; (2) prices move countercylically, falling at seasonal quantity peaks; and (3), supply shocks, in the form of price movements in agricultural inputs, explain few, if any, of the seasonal retail price movements. The paper adds to empirical research that uses micro level price data to summarize and evaluate firms' pricing responses to seasonal and product lifecycle demand shocks (Warner and Barsky [1995], Pashigian and Bowen [1991], Borenstein and Shepard [1996]).

I next describe the underlying dataset, which is based on electronic supermarket scanner information, and summarize the methods used to define seasonal products and to select seasonal peaks. The following section documents the primary empirical observation of the paper, extensive peak period price declines. I then provide evidence from several sources to show that cost shocks cannot account for the price movements. Absent declines in marginal costs, we should not see countercyclical price moves in perfectly competitive markets, but we can if markets are imperfectly competitive. The final section provides supporting evidence for two related models of intertemporal pricing under imperfect competition.

## II. MICROLEVEL FOOD PRICE DATA

The analysis relies on supermarket prices collected by the A. C. Nielsen Company and marketed as the Nielsen ScanTrack Database (NSD). NSD files are derived from the data read into electronic scanners at supermarket checkout counters. The raw data, contained in an item's 'bar code', include detailed product information on brand, flavor, packaging type, and container size. The scanner also registers each item's retail price, exclusive of coupon discounts.

Supermarkets provide Nielsen with summary scanner data on quantity and sales, by item, as well as some information on item and brand promotion activities. Nielsen aggregates the data from a nationwide sample of 3,000 supermarkets; this paper's dataset includes nationwide estimates of monthly quantities and dollar volumes for each item, brand, and product class. NSD prices are 'unit values', estimated national sales volume divided by estimated national quantity.

Since scanner data rely on bar codes, products sold in variable weights (produce, fresh meats) are not covered. Covered items are organized into 613 product classes in grocery, refrigerated, frozen, delicatessen, and packaged meat categories. Each NSD product class includes brands, and items sold under each brand name. For example, Star-Kist is a well-known *brand* in the canned tuna *product class*; Star-Kist canned tuna *items* are in turn differentiated by type (such as chunk light), container size (6.25 ounce is the most common), and packing medium (packed in water, or in oil). The 613 product classes include over 220,000 separate items.

We face two important issues of sample design. First, we must identify a set of seasonal food products. Second, we must define the appropriate level of product aggregation for analyzing seasonal effects on prices—product classes, brands, or items. Start with aggregation. Average NSD brand and product class prices are sensitive to changes in the item composition of those aggregates; that is, average class and brand prices can fluctuate even when no individual prices change, solely because of changes in the mix of package sizes or flavors that make up those aggregates. In order to emphasize actual price changes, the analysis emphasizes prices and quantities at the most detailed level of aggregation—identical items aggregated across outlets. In order to emphasize price movements among representative, nationally distributed items with lengthy price histories, I use prices for the largest selling item of the largest selling brand in the selected NSD product classes.

Next we must select seasonal products from among the NSD product classes. Monthly sales patterns for each product class were reviewed and those with visual evidence of seasonality were set aside for closer evaluation. Regressions of the following form were run for each item, to select seasonal products and to identify their peak, shoulder, and off-peak months:

$$(1) \qquad q = \alpha + \sum \beta^* Y + \sum \gamma^* M + \mu$$

where $q$ is the natural logarithm of monthly item quantity, $Y$ is a vector of dummy variables for each year in the sample, and $M$ is a vector of dummy variables for each month. The data cover an 81 month period from April 1988 through December 1994.[1]

Peak months are those with the highest $\gamma$ value in Equation (1). I then estimated 'peak effect' regressions for each item:

$$(2) \qquad q = \alpha + \sum \beta^* Y + \gamma^* PEAK + \eta^* SHOULDER + \mu$$

where $PEAK$ is a dummy variable for the peak month selected in regression (1) and $SHOULDER$ is a dummy variable (sometimes two) for other high sales months.

Equation (2) was used to select the final sample; the goal was to develop an adequately sized sample of products with strong and clear seasonal quantity movements, while excluding products with less distinct quantity cycles. Two criteria drove sample selection. First, the coefficient $\gamma$ on $PEAK$ had to predict quantity increases of at least 35% over the off-peak average. That criterion allowed for a reasonable sample size, while excluding a number of products with less distinct quantity shifts. Second, shoulder months had to be either adjacent to the peak month, or they could form a double peak (for example, December and April for products whose sales surge at Christmas and Easter). Items with at least three nonadjacent peaks were rejected, since such movements likely reflect replacement patterns for nonperishables and not seasonal shifts. Using these criteria, the data set consists of the leading item of the leading brand in 48 different NSD product classes.

The appendix lists product classes, peak months, and consumption categories. These items have very strong seasonal shifts, with a mean peak quantity increase of almost 200%, and the set corresponds closely to *a priori* notions of seasonal products. Ten are picnic and outdoor grilling

[1] NSD reporting periods end in the middle of calendar months—'June' in the NSD files begins in mid-May and ends in mid-June. Because NSD months have variable lengths (eight four week months and four 5 week months), I standardized all quantity flows to 30 day rates.

products with strong summer peaks, including bratwurst, frankfurters, ketchup, barbecue sauce, ice cream cones and bars, and various cold drinks (liquid ice pops, powdered soft drinks, tea mixes). Three peaks are associated with seasonal religious practices (two processed fish products purchased during Lent and matzo meal, with October and March peaks), while only one (apple cider) occurs at a traditional fall harvest time. Thirty-one products peak in December and January, and 17 of them cover baking ingredients (such as baking chocolate, baking cocoa, flour, or gingerbread mixes). Nine other December-January peaks are for products typically consumed as condiments and accompaniments at holiday gatherings (yams, chili sauce, sour cream, sweet pickles). Finally, eight products, including the remaining five with December-January peaks, are product typically consumed in cold weather months, either directly (hot cereal) or as ingredients (canned tomatoes).

## III. SEASONAL PRICING REGULARITIES

The first step in the empirical analysis is to ascertain the behavior of product prices at seasonal peaks. I ran a series of Equation (2) regressions, with item price replacing quantity:

$$(3) \qquad p = \alpha + \sum \beta^{*} Y + \gamma^{*} QPEAK + \eta^{*} QSHOULDER + \mu$$

where $p$ is the log of item price, $Y$ is a vector of dummy variables for each year, $QPEAK$ is a dummy variable for the seasonal quantity peak month, and $QSHOULDER$ is a dummy variable (two variables for some products) for shoulder months. The specification tests for differences in the level of prices in the peak month, relative to mean off-peak prices.

Table I shows summary results from the 48 regressions, while the appendix reports estimated peak period price and quantity effects for each product class. The evidence shows widespread and persistent price declines at seasonal quantity peaks—figure 1 is clearly not anomalous, but rather is typical. No item price rose by a statistically significant amount in the peak month, while 44 of 48 fell by statistically significant amounts. The middle panel of Table I summarizes the distribution of t statistics for the null hypothesis that the coefficient on $PEAK$ was zero (recall that each price regression has 81 monthly observations, and 71 or 72 degrees of freedom, depending on the number of shoulder months). Seventeen of the 44 'significant' coefficients had $t$ statistics in excess of 6, while another 13 were above 4. Only 2 were below 2.5 (individual results are again reported in the Appendix).

On average, quantity rose by 199% and price fell by 7.5% in the peak month, relative to off-peak monthly means. Seven products had mean peak price declines below 3% while 6 had mean declines exceeding 15%. Prices

TABLE I
PEAK PERIOD PRICE CHANGES FOR 48 SEASONAL FOOD PRODUCTS

| Direction of Peak Price Change | Number of Products |
|---|---|
| Rise | 0 |
| Fall | 44 |
| No change | 4 |
| **Distribution of *t* Statistics** | **Number of Products** |
| 0–2 | 4 |
| 2–4 | 13 |
| 4–6 | 14 |
| >6 | 17 |
| **Mean Changes at Peak** | **Percent** |
| Price, relative to offpeak mean | −7.5 |
| Price, relative 6 months earlier | −8.5 |
| Price, relative to peak price month | −10.5 |
| Quantity | +199 |
| Price-adjusted quantity | +87 |

Notes: Peak Price Change estimates are derived from regressions described in text. 'No Change' refers to no statistically significant price change at peak period. Individual regression results are described in Appendix A.

fell by 10.5%, on average, when compared to prices in the highest month, and by 8.5% when compared to prices 6 months prior to the peak. Peak period movements were a principal source of sample price fluctuation: the mean average increase in prices among the 48 products was only 1.4% per year over the 1987–1994 period.[2]

### *Are the Findings Robust and Seasonal?*

The findings described in Table I are striking for their size and for their pervasiveness. But there are specific reasons why the findings could be in error, or could be dependent on a particular product definition, or could simply be unremarkable. I consider each in turn.

First, they could reflect errors in measuring prices. Recall that NSD prices are unit values, nationwide item sales divided by nationwide item quantities. Some unit value measures, whose sales and quantity estimates are derived from different sources, are clearly prone to measurement errors that introduce spurious inverse price and quantity correlations: given sales, quantity overestimates will result in price underestimates and quantity

[2] The inflation rate is typical for NSD products; while much lower than the Consumer Price Index for Food at Home (3.7% annually), NSD price trends match food price trends reported in Reinsdorf [1994]. Reinsdorf and Moulton [1997] show why a fixed-weight CPI index yields higher rates of price increase than the unit value measure embodied in the Nielsen data.

underestimates will result in price overestimates. But measurement errors of this sort should not arise in scanner data.[3]

Now consider the choice of product definition. The sample consists of observations on the largest selling item of the largest selling brand in each selected product class. Item-based data offer cleaner measures of price change; average brand prices can change because the item composition of brands change, even if individual item prices are unchanged. But the results are insensitive to the choice of aggregation. The analyses of Table I were rerun using brand level prices, calculated as price per ounce across all items for the largest selling brand in a class. Forty two brands (instead of 44 items) had statistically significant declines in price, and the average price decline was 7%, instead of a 7.5%. Given the noise introduced into brand level calculations by shifts in the item composition of brands, the results are remarkably close.

Third, it's possible that the findings are not as remarkable as they first appear. Recall that nearly two thirds (31) of the sample's quantity peaks occur in December and January. Suppose all food prices generally display seasonal price declines at that time of the year. In that case, the results in Table I could be driven by general food price trends, and the apparent association between seasonal quantities and prices could then be spurious.

Aggregate food price seasonals were checked using monthly values of the Consumer Price Index for Food at Home for the period 1985–1994. January is the only month with a statistically significant price effect, but January prices increase by 1.28% relative to CPI trends.[4] That is, the January CPI shows persistent small seasonal price increases when seasonal product prices are falling (December also shows average annual increases, not significant, of 0.34%).

Finally, there is a question of endogeneity—the selected seasonal quantity peaks could simply reflect demand responses to price cuts, instead of seasonal shifts in demand. While it seems unlikely that prices account for all of the observed seasonal quantity increases—that, in other words, people would bake dramatically more in the summer if only prices were a

[3] Scanner quantities can be mismeasured if a barcode is misread at the checkout counter, if the manufacturer enters the wrong code on a container, or if information systems incorrectly match barcodes and items. These quantity errors can be positive or negative, and are unlikely to be correlated with the seasonal cycle. More important, when quantities are reported with error, sales are reported with a matching error. If miscoding causes aggregate quantity to be understated, then aggregate sales will also be understated, and unit values will be unaffected.

[4] The results come from a model in which the monthly log change in the Food at Home CPI was regressed on annual and monthly dummy variables. The only significant coefficient was that on the month of January, suggesting persistent relatively large CPI jumps in January.

little lower—it is still useful to try to sort out the separate effects of seasonal price declines on seasonal quantities.

In order to investigate the likely price effects, I reran Equation (2) with the log of item price included. Since contemporaneous prices are likely to be correlated with the error term in Equation (2), I used an instrumental variable for price (predicted values from a regression on year and month dummies as well as three and six month lags in prices—one month lags will be correlated with contemporaneous error terms, if temporary price cuts lead to intertemporal substitution). The mean estimated price elasticity over the 48 regressions was $-6.36$, with an interquartile range of $-3.28$ to $-8.88$, quite reasonable for monthly brand level price elasticities.[5] All but four were statistically significant. Elasticities like this, combined with persistent seasonal price declines, have important estimated effects on seasonal quantities. For example, using the mean price elasticity and peak period price decline, the peak period price effect will account for one third of the seasonal quantity increase, cutting the mean seasonal effect from a 199% increase over mean offpeak quantities (Table I) to a 134% increase (and an 87% increase over quantities predicted on the basis of low peak prices alone).

This simple model provides very conservative estimates of seasonal quantity shifts, because it includes the leading brand's price but omits prices for private label and competing branded products (as well as substitute product classes).[6] Since private label and competing brand prices also decline at seasonal peaks, and since those declines should reduce leading brand quantity, coefficients on *PEAK* should be biased downward.[7]

Seasonal price cuts have important estimated effects on sales. But even when estimated with a method likely to lead to substantial downward biases in estimated seasonal effects, most products still show large seasonal demand shifts, independent of price cuts. Moreover, systematic seasonal price cuts must still be explained. I argue below that marginal costs don't

[5] Substitution can come from closely competing brands and from competing product classes. In monthly data, there's also considerable scope for intertemporal substitution, as sales lead to increased sales this month and reduced sales next month.

[6] This particular dataset provides extensive temporal and product detail, but no cross sectional geographic detail. Brand prices are highly collinear with one another in datasets that lack cross sectional elements. Scanner data that include cross sectional observations on stores or cities may be better suited to demand estimation because of cross sectional price and demographic variation. Scanner data purchase prices increase quite sharply as the purchaser adds more product, temporal or geographic detail, so buyers usually economize on at least one dimension.

[7] Moreover, seasonal price declines are virtually the only source of price variation for a few products; in those cases, estimated price elasticities are quite large (that for bratwurst was $-22.3$), and one can't reliably disentangle seasonal demand shifts from seasonal demand responses to price cuts. Indeed, the coefficient on *PEAK* became statistically insignificant for 3 products with limited offpeak price variation (bratwurst, hot dogs, and refrigerated pies).

appear to be falling at peak seasons; without cost shocks, we need to find another reason for firms to cut prices at seasonal peaks.

## IV. DO COMMODITY PRICES DRIVE RETAIL PRICE SEASONALITY?

Retail food prices fall during seasonal demand surges. Do the declines reflect declines in materials costs? The evidence, drawn from two sources and summarized below, strongly suggests that the answer is no, and that farm to retail margins narrow sharply at seasonal peaks.

In competitive markets, prices could fall in the face of seasonal demand surges, if there were coincident positive supply shocks. The supply shocks would not even have to be particularly large, as long as market supply curves were relatively elastic with respect to price, while market demand curves were relatively inelastic. Persistent seasonal supply shocks are unlikely be driven by seasonal declines in wages or capital costs, but they may result from seasonal price declines for key agricultural inputs. Seasonal demand surges could be correlated with supply shocks if current patterns of seasonal food consumption (for example, baking at Christmas) reflect traditional harvest celebration periods in earlier cultures.

Most of these products are processed from a few key agricultural commodities, which can make up significant shares of total cost. For example, wheat flour is the primary ingredient in nine products, while manufacturing grade fluid milk is used in five, cocoa in four, and vegetables grown for processors (tomatoes, yams, cucumbers) are used in seven. Agricultural input costs account for large shares (25% to over 50%) of processors' value of shipments for more than half of the products, based on published 1992 Census of Manufactures data for the associated four digit industries.

Do the component commodities display seasonal price fluctuations? Table II reports the results of testing for systematic seasonal price fluctuations for a set of key agricultural commodities.[8] The table lists the month of the seasonal price trough for each commodity, as well as the mean off-trough to trough price decline. Most of the commodities display statistically significant seasonal price fluctuations, and some are rather large, although the mean commodity price decline is less than half the mean retail price decline.

But while commodity prices do show some seasonality, they don't match retail product price trends very well, for three reasons. First, some commodities with seasonal price movements are ingredients in several

[8] I followed the same strategy used to identify and estimate price and quantity peak effects for retail products. The commodity price regressions typically covered monthly observations over a ten year period from January 1985, through December 1994. Data sources are described in the JIE editorial website.

TABLE II
SEASONAL LINKS BETWEEN AGRICULTURAL AND RETAIL PRICES

| Agricultural Commodity | Month(s) of Price Trough | Mean % Seasonal Price Decline | Number of Retail Products | Number with Matched Troughs |
|---|---|---|---|---|
| Cocoa | None | None | 2 | 0 |
| Corn Starch | Oct-Dec | 14.2 | 2 | 2 |
| Contract Tomatoes | Jul-Aug | 3.0 | 4 | 1 |
| Contract Veg. (other) | None | None | 3 | 0 |
| Milk (Manufacturing) | Jun/Jul | 4.0 | 5 | 1 |
| Oats | Jul-Aug | 11.8 | 1 | 0 |
| Strawberries (Contract) | None | None | 1 | 0 |
| Sugar | Oct-Dec | 1.0 | 8 | 2 |
| Tea | Jun-Aug | 11.6 | 1 | 1 |
| Wheat | Jun/Aug | 10.8 | 9 | 1 |
| Total | | 3.36 | 36 | 8 |

Note: Sources for commodity prices are listed in JIE editorial website.

retail products with different seasonal peaks. For example, manufacturing grade milk has a clear summer price trough, and is a key input in four retail products: ice cream bars, cream cheese, dairy dip, and sour cream. But only one, ice cream bars, has a summer quantity peak, when the retail price trough matches a commodity price trough. The other three retail products peak in winter, with seasonal lows in retail prices that coincide with commodity price peaks. Similarly, tomatoes are grown for processors under annual contracts that contain adjustments providing higher prices for 'late season' deliveries, in order to compensate growers for higher production risks in the early fall. Hence, contract tomato prices show some seasonal fluctuation, and are lower early in the summer growing season. But only one of four relevant retail products has a seasonal demand peak that matches the commodity price trough, and it has a seasonal retail price decline six times greater than the commodity price decline. The other three retail product demand peaks occur out of the growing season and presumably cause processors to incur higher commodity costs, for storage of processed tomato products.

Second, some commodities show reasonably strong price patterns, but seasonal commodity price trends move in the opposite direction from related retail product prices. For example, wheat prices peak in midwinter and fall in summer. But flour and most of the other retail products that use wheat flour have late fall seasonal quantity peaks (and retail price declines), when wheat prices are rising. Similarly, hog prices peak in summer, when bratwurst and hot dog prices fall sharply.

Third, some show no seasonal effects. Some fruits and vegetables (strawberries, pickles, yams) are typically produced under annual contracts with no contractual price adjustments for seasons. Moreover, cocoa, an important ingredient in several retail products with strong seasonal price movements, shows no evidence of seasonal price fluctuations.

Some commodity prices (such as tea and corn starch) do match seasonal retail product price trends. But on the whole, there is little evidence of systematic linkages. The commodities listed in Table II are the primary ingredient for 36 of the sample's 48 retail products. The final column of the table lists the number of retail products whose seasonal price declines matched a seasonal commodity price decline. I apply the idea of a match generously: seasonal retail and commodity price movements are said to match if the commodity trough occurs within 4 months prior to the retail trough, and they are said to match even if the retail price decline substantially exceeds the commodity decline. Even under this definition, there is as much evidence for commodity price increases at seasonal quantity peaks as there is for commodity price declines, and only eight of the 36 products have commodity declines that match retail product declines.

There's a second way to investigate the possible effects of commodity prices: Table III compares seasonal price movements for the sample and for corresponding 'private label' items, which are sold under retailer brand names and are manufactured by the retailer or, more often, for the retailer by an independent manufacturer. Private label products are sold in 41 of the 48 product classes in the sample. Seasonal commodity price shocks ought to affect all brands in a product category; in particular, there is no reason to believe that commodity costs account for a larger share of total costs for leading brands than for private label brands and items in a category. If that is true, then changes in commodity prices should not lead to changes in *relative* branded-private label prices.

Table III
Matching Branded and Private Label Products: Peak Period Price Changes

| Directions of Price Change at Peak | | | |
|---|---|---|---|
| Corresponding Private Label Product | Branded Product | | |
| | Falls | No Change | Total |
| | | —Number of Items— | |
| No Change | 12 | 3 | 15 |
| Falls Less | 16 | 0 | 16 |
| Falls as Much | 5 | 0 | 5 |
| Falls More | 5 | 0 | 5 |
| Total | 38 | 3 | 41 |

Note: Forty one of the 48 seasonal product categories have private label brands. 'Directions of Price Change' all refer to statistically significant price effects; for example, 'falls less' means that the private label products have statistically significant price declines at the peak month, and that the difference between private label and branded price declines is statistically significant.

In order to compare leading brand and private label peak period pricing, I first ran regressions based on Equation (3) for the 41 private label prices, and then for the ratio of branded to private label prices in each product class. Table III reports that peak period private label prices fell, by statistically significant amounts, in 26 of 41 product classes. By itself, that statistic suggests that commodity prices could play some role. But leading brand prices fell, when compared to private label prices rather than the leading brand's annual average, in 28 out of 41 cases, while increasing in 5, and remaining the same in 8.[9] The average change in leading brand prices, relative to private label, was 6.2%; the relative decline was only slightly less than the absolute decline noted in Table I. Commodity prices are unlikely to be a dominant explanation when branded prices decline relative to the prices of competing private label products.

Other cost effects are quite unlikely. These aren't natural monopolies, so output increases should not lead to falling marginal costs. Persistent output growth could lead to reduced costs through learning effects or through external economies, but seasonal demand increases shouldn't lead to those long-run cost effects. The data do not support the hypothesis that peak period retail price declines reflect seasonal declines in costs. Rather, seasonal quantity movements appear to reflect seasonal demand fluctuations, and the evidence suggests that retail prices decline in the face of seasonal increases in demand.

## V. WHY CUT PRICES WHEN DEMAND INCREASES?

Much recent interest attends cyclical pricing behavior, with particular attention paid to models of imperfect competition in which oligopolists' prices vary countercyclically. In their survey of that literature, Rotemberg and Woodford [1999] advance four reasons why markups (prices, given marginal costs) might vary countercyclically. Two, tacit collusion and changes in price elasticities of demand, seem relevant for seasonal cycles.[10] In the first case, tacitly collusive outcomes are supported by repeated interaction among the players. Firms collude in the sense that they have an implicit agreement that deviations from a collective understanding on

[9] We can easily reject the hypothesis of equal peak period price changes between private label and branded products, at a 0.001 significance level in a Pearson chi-square test for goodness of fit.

[10] Two others, customer markets with switching costs and procyclical supplier entry, apply poorly to seasonal cycles. If new buyers purchase during demand expansions, and if initial brand choices influence later choices because of customer switching costs, then firms may aim to attract new buyers by cutting prices during seasonal expansions. But switching costs are unlikely to be important in this sample (chocolate syrup involves no compatibility or brand-specific learning effects), and seasonal demand flows largely reflect intermittent rather than new buyers.

prices will be punished; there are not necessarily any explicit, and illegal, agreements.

In these models, self-enforcing collusion depends on the current gain from defection being smaller than the anticipated future loss triggered by the defection. Gains occur when the firm cuts prices below the collusive price and attracts new sales away from rivals, while losses arise in the future as firms react to defection by cutting prices. Rotemberg and Woodford [1999] show that the current gain to defection increases at high demand periods, such as seasonal peaks, and so therefore does the incentive to defect, because defecting firms will attract more buyers. Reactions will also be less effective because they will occur at low demand periods; hence the present value of future losses will be lower. Implicitly collusive prices can be sustained only by reducing the gains from cheating at the high-demand period, and that can be done by reducing the collusive price at that period. As a result, optimal collusive prices fall during demand surges. Moreover, Rotemberg and Woodford argue that high expected future demand should raise current markups, because high future demand raises the costs to current defection.

The second case considers systematic seasonal increases in price elasticities of demand. The acceptance of this well-known possibility has been limited, in Rotemberg and Woodford's words, by a lack of persuasive reasons for price elasticities to vary with demand. Bils [1989] reintroduced the idea in a paper in which he argued that new customers, with weaker brand attachments, begin purchasing during booms and that this compositional effect makes demand more price-sensitive. However, he could provide no direct evidence that buyer composition does in fact change during booms.

But retail products have one potential source of seasonally increasing demand elasticities—'thick market' effects that reduce the costs of information acquisition for buyers. In Warner and Barsky's [1995] analysis of retail pricing patterns for durable goods in the Christmas shopping season, they noted a widespread use of markdowns before Christmas and a strong tendency for markdowns to occur on weekends rather than during the week; in short, prices were cut during the periods of greatest demand. They argue that increased customer flows at those periods (to shopping centers to purchase multiple products) provided sellers with incentives to initiate or to increase information expenditures, which reduce households' costs of acquiring product information. Reduced information costs in turn allow for easier comparison across brands and higher cross and own price elasticities.

Warner and Barsky rely on Grossman and Shapiro's [1984] model of informative advertising in imperfectly competitive markets. In that model, reductions in the per customer cost of advertising lead firms to increase promotional expenditures; increases in informative advertising lead in turn to increases in market information, firm demand curves that are more price-elastic, and reduced prices.

Can seasonal demand surges lead to reduced per-customer costs of informative advertising? Food product information is typically conveyed in newspaper advertisements, and the cost of an ad varies with the circulation of the newspaper and the size of the ad. During seasonal demand surges, a higher proportion of newspaper readers will be prospective purchasers of a product. With unchanged circulation, the per reader price of an advertisement will be unchanged, but the per customer price will fall since more readers will be customers.

The two reasons are not mutually exclusive. Defections from collusive pricing should yield higher current gains if buyers are more informed of, and responsive to, price cuts. In turn, future losses from retaliatory price cuts will be smaller if buyers are less informed of price cuts. As a result, information induced variations in firm-specific demand elasticities should reinforce models of tacitly collusive pricing.

NSD files offer some interesting evidence with regard to these models. Of course, prices fall at seasonal demand peaks, and they peak during seasonal demand troughs, when demand is virtually certain to rise in the future. But seller concentration also has a strong relationship with seasonal price-cutting, in a manner consistent with models of tacit collusion. Consider Table IV, which sorts the sample into quartiles ordered by 1995 three firm concentration (the share of product class sales held by the three largest firms in the class). Mean CR3 in the highest quartile (Q1) is 92, and markets in this quartile are typically dominated by a single strong brand—the mean leading brand market share is 67%. If seasonal

TABLE IV
CONCENTRATION AND SEASONAL PRICE CHANGE

| | Concentration Quartiles | | | |
|---|---|---|---|---|
| | Q1 | Q2 | Q3 | Q4 |
| *Market Structure Means* | | | | |
| 3 firm concentration | 92 | 80 | 63 | 40 |
| Largest brand's share | 67 | 58 | 33 | 24 |
| Private label share | 2 | 11 | 13 | 19 |
| *Peak Price Change Statistics* | | | | |
| Mean Price Decline at Peak | 3.4% | 6.9% | 12.4% | 9.2% |
| Standard Deviation | 2.71 | 2.84 | 8.36 | 5.42 |
| *t* test for difference from Q1 | | 2.44 | 3.94 | 3.29 |
| *t* test for difference from Q2 | | | 2.18 | 1.86 |
| *t* test for difference from Q3 | | | | 1.32 |

Note: Sample of 48 products was divided into four concentration quartiles of 12 products each. Mean price decline at peak is mean of twelve estimates from peak period price regressions described in text.

cycles reflect implicitly collusive pricing among rival sellers of substitute brands, then seasonal price effects should be relatively weak in this quartile. Indeed, they are—the mean peak period price decline in Q1 is 3.4%, less than half of the sample average.

Now consider the two lowest CR3 quartiles (Q3 and Q4), which usually have at least two, and often three, strong brands as well as private label competition. The mean leading brand market share in Q3 is half that of Q1, while private label sales shares are 13% and 19% in Q3 and Q4, but only 2% in Q1 (Table IV). Oligopolistic interaction should be more relevant in Q3 and Q4 product classes. Indeed, peak period price declines are 3 to 4 times larger than in Q1, with the differences in means being highly significant. Countercyclical seasonal pricing is clearly stronger in markets with competing strong brands, where oligopolistic rivalry should matter.[11]

Now consider some evidence on information. I argued above that per customer promotional costs should fall at seasonal peaks, and the evidence shows large increases in promotional activities at peaks.[12] NSD files record four types of retail promotions: in-store displays, major newspaper ads, coupon ads, and line (minor) ads. I use measures of promotion intensity—the share of a product's market in which a particular type of promotion is appearing.[13] To identify seasonal promotion patterns, I ran the peak effect regressions outlined above in Equation (2), but with promotion intensities replacing quantities. Table V, which reports summary data on seasonal variations in promotion for the 48 products in the sample, reveals powerful seasonal fluctuations. Consider the first row, on in-store displays. On average across the 48 products, displays are set up in 7.7% of the product's potential market in off-peak months, but the mean intensity surges to 25.7% of the potential market in peak months. All 48 products show statistically significant increases in display intensity in the peak month. Most products have statistically significant increases in major ads and line (minor) ads at seasonal peaks, but those are relatively minor promotional tools for these products, and products that do not show

[11] Analyses of tacitly collusive pricing sometimes test the models by relating measures of concentration to countercyclical pricing (see Rotemberg and Woodford's summary). But concentration may be a poor measure of the potential for tacit collusion, if high concentration reflects dominant firms and not oligopolies.

[12] Bils [1989] also finds that advertising is strongly procyclical with respect to movements in the business cycle; he argues that advertising is directed primarily at new customers and that booms therefore reflect the entry of new customers.

[13] The NSD file reports the sales-weighted share of sample stores selling an item, and the sales-weighted share of sample stores carrying a particular promotion for an item. Divide the second share by the first to obtain the share of the product's market that is having a promotion. For example, if 50% of sample supermarkets (weighted by volume) sell an item, and if 18% are offering coupons for the item, then promotions are offered in 36% (0.18/0.50) of the item's market.

TABLE V
SEASONAL SHIFTS IN PROMOTIONAL ACTIVITIES

| Promotional Activity | Mean Incidence of Activity (%) Off-peak | Peak | # of products with significant peak increases |
|---|---|---|---|
| Displays | 7.7 | 25.7 | 48 |
| Major Ads | 3.0 | 6.2 | 33 |
| Coupon Ads | 12.5 | 30.2 | 44 |
| Line Ads | 1.5 | 6.2 | 41 |

Note: The measure of incidence is the percent of a product's market in which the activity was carried out in at least one week of the month.

increases in major ads typically do not run major ads at any time. The striking news concerns coupon ads: 44 of the 48 products show statistically significant increases in coupon ads in peak months, and the increase in mean coupon intensity is quite large, from 12.5% of the potential market to 30.2%. The surge in coupons also suggests that Table I understates typical seasonal price declines, since scanner data show shelf prices, not net of coupon prices. If newspaper ads and displays are informational forms of advertising, then it appears that price and product information are more widely available at seasonal peaks, and that such thick market effects may help account for seasonal price declines.[14]

## VI. CONCLUSIONS

I've used scanner data to establish four strong empirical regularities in a sample of retail food products with strong seasonal demand swings. Prices move counter-cyclically; falling at seasonal demand peaks. Price declines do not appear to be linked to declines in input prices. Indeed, there is as much evidence for input price increases during periods of peak product demands; in short, margins appear to fall at seasonal peaks. Peak price declines are strongly related to market concentration: declines are much larger in markets with competing brands than in markets with a single dominant brand. Finally, price declines coincide with surges in promotional activity at seasonal peaks.

The empirical regularities are not consistent with perfect competition,

[14] Since the NSD records only retail prices, we cannot be sure whether manufacturers or retailers are cutting peak prices. Three bits of evidence point to some role for manufacturers. First, coupon sales in the NSD are manufacturer coupons, and Table V shows that coupon offerings increase sharply at peaks. Second, manufacturer concentration shows a distinct association with the size of peak period cuts. Finally, I've interviewed food wholesalers and manufacturers in connection with another project, and have been told that manufacturers' prices for many of these products are frequently cut at peak demand periods. The evidence does not, of course exclude, additional retailer cuts.

because demand increases, absent cost shocks, should increase prices in competitive markets. But they are consistent with models of oligopoly rivalry and 'thick market' effects. Seasonal demand increases reduce the effective costs of informative advertising, and increased informative advertising by retailers and manufacturers in turn allows for increased market information and greater price sensitivity on the part of buyers. Faced with greater but more price elastic demands, sellers facing rival brands cut prices.

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Appendix Table A
Product Classes, Peaks, and Peak Quantity and Price Changes

| Product Class | Peak | Quantity Change (%) | Price Change (%) | $t$ statistic on Price Change |
|---|---|---|---|---|
| 1. Baking Chips (b) | Dec | 291 | −6.0 | 7.07 |
| 2. Baking Chocolate (b) | Dec | 179 | −6.0 | 6.33 |
| 3. Baking Powder (b) | Dec | 131 | −1.5 | 2.80 |
| 4. Barbecue Sauce (s) | Jul | 160 | −17.2 | 5.31 |
| 5. Bratwurst (s) | Jul | 533 | −11.7 | 3.59 |
| 6. Brown Sugar (b) | Dec | 234 | −10.6 | 8.24 |
| 7. Butter (b) | Dec | 94 | −8.0 | 5.63 |
| 8. Chili Sauce (c) | Jan | 143 | −5.2 | 6.18 |
| 9. Chocolate Syrup (o) | Jun | 40 | −2.7 | 5.89 |
| 10. Cider, Apple (h) | Nov | 837 | −7.3 | 3.37 |
| 11. Cocoa (b) | Dec | 131 | −2.5 | 11.83 |
| 12. Coconut (b) | Dec | 497 | −19.7 | 10.99 |
| 13. Confectionery Paste (b) | Jan | 391 | −3.5 | 3.07 |
| 14. Cookie Mixes, ref. (b) | Dec | 77 | −4.9 | 6.41 |
| 15. Corn Starch (b) | Dec | 62 | −1.9 | 2.08 |
| 16. Cream Cheese (c) | Jan | 151 | −6.7 | 6.05 |
| 17. Dairy Dip (c) | Jan | 43 | −5.9 | 8.26 |
| 18. Dates (c) | Dec | 274 | −2.1 | 3.86 |
| 19. Dinner Rolls, ref. (c) | Jan | 101 | −5.8 | 7.03 |
| 20. Dry Yeast (b) | Dec | 121 | −0.7 | 0.77 |
| 21. Fish, breaded, fzn. (r) | Mar | 69 | −8.4 | 6.89 |
| 22. Flour, all-purpose (b) | Dec | 210 | −17.8 | 5.78 |
| 23. Frankfurters (s) | Jul | 116 | −18.1 | 10.26 |
| 24. Fruit Drinks, fzn. (s) | Jul | 224 | −8.1 | 6.73 |
| 25. Gingerbread Mix (b) | Dec | 284 | −5.4 | 3.63 |
| 26. Gravy, canned (c) | Dec | 96 | −6.9 | 4.48 |
| 27. Hot cereal (o) | Jan | 80 | −5.6 | 3.79 |
| 28. Ice cream bars (s) | Aug | 89 | −7.2 | 6.23 |
| 29. Ice cream cones (s) | Jul | 180 | −5.7 | 3.18 |
| 30. Jam (o) | Feb | 58 | −11.0 | 5.24 |
| 31. Ketchup (s) | Jun | 62 | −8.7 | 4.81 |
| 32. Liquid ice pops (s) | Jul | 433 | −5.8 | 5.99 |
| 33. Matzo meal/mixes (r) | Apr | 479 | −5.4 | 3.56 |
| 34. Milk/water additives (o) | Jan | 517 | −13.3 | 3.74 |
| 35. Molasses (b) | Dec | 155 | −2.1 | 5.16 |
| 36. Muffin mixes (b) | Jan | 40 | −2.4 | 2.94 |
| 37. Pie shells, ref. (b) | Dec | 166 | −7.2 | 5.67 |
| 38. Pies, fzn. (o) | Dec | 141 | −10.0 | 7.71 |
| 39. Powdered soft drinks (s) | Jul | 78 | −6.5 | 4.67 |
| 40. Pudding Mix (o) | Dec | 68 | −3.1 | 2.71 |
| 41. Salmon, canned (r) | Mar | 79 | −6.2 | 2.12 |
| 42. Shortening (b) | Dec | 53 | −2.0 | 1.31 |
| 43. Soup, canned (o) | Dec | 53 | +1.4 | 1.17 |
| 44. Sour Cream (c) | Jan | 91 | −7.8 | 4.74 |
| 45. Sweet Pickles (c) | Jan | 132 | −17.1 | 6.94 |
| 46. Tea mixes (s) | Jul | 216 | −13.2 | 4.62 |
| 47. Tomatoes, canned (o) | Feb | 51 | −0.8 | 1.18 |
| 48. Yams, canned (c) | Dec | 867 | −23.9 | 12.53 |

Note: Letters in parentheses refer to consumption categories—baking ingredients (b), religious influences (r), harvest periods (h), holiday meal condiments and accompaniments (c), summer outdoor grilling and picnic items (s), and other products (o). Quantity and price changes refer to peak period changes relative to offpeak means, and are derived from $\gamma$ coefficients in Equations (2) and (3). Last column reports $t$ statistics for hypothesis that $\gamma$ in Equation (3) equals zero.

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